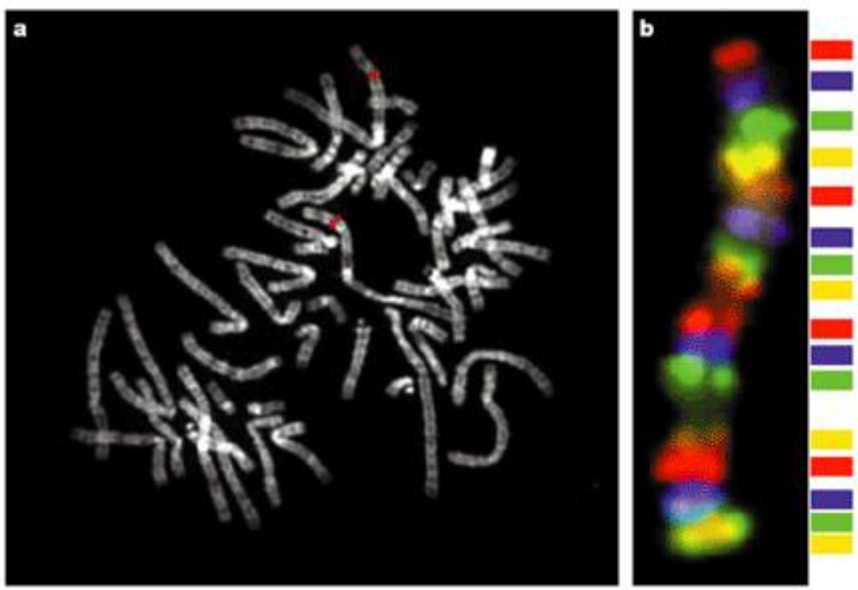
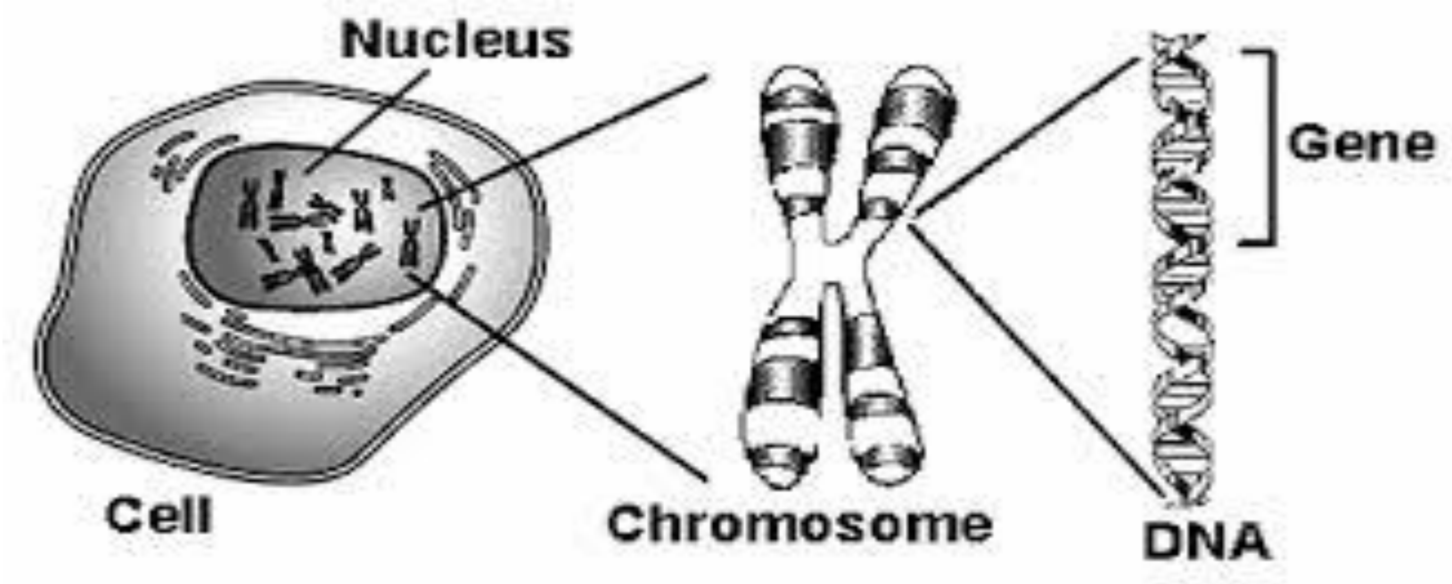
Cytogenetics



Nature Reviews | Genetics

Chromosomes



A thread like linear strand
of DNA associate with protein in the nucleus
carry the genes
Function in transmission of
hereditary information

Cell Cycle

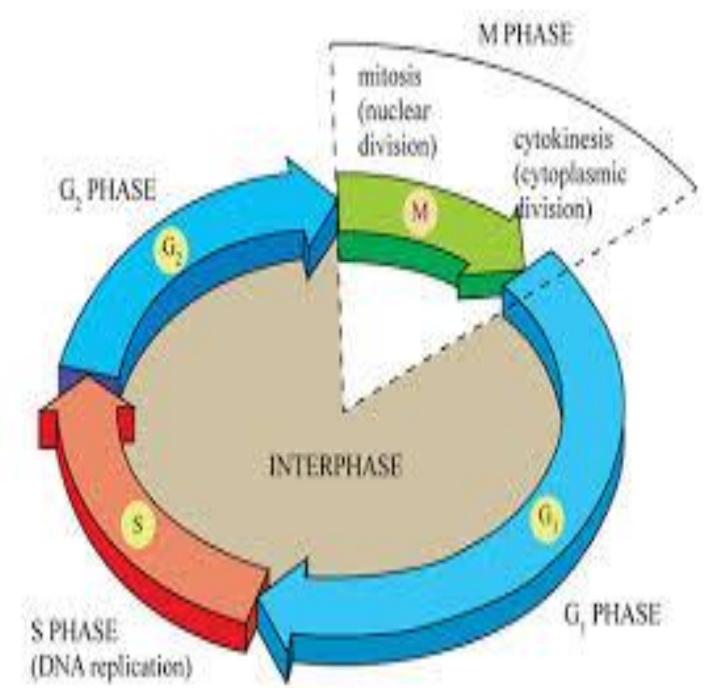
A growing cell undergoes a cell cycle that is consists of two periods;

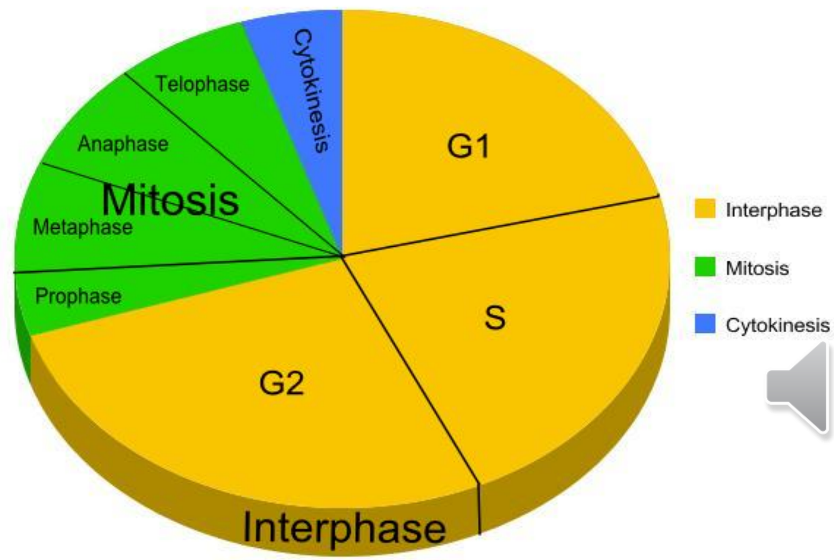
(1) **Interphase** (the period of non apparent division), and

(2) The period of **division**.

In eukaryotes, division generally takes place by mitosis or meiosis.

Division. Interphase Division.





Interphase

The cycle is divided into four successive intervals:
G_1, S phase, G_2 and mitosis

Interphase

First growth stage (G_1) is the period between the end of mitosis and the start of DNA synthesis,

G1 stage characterized by:

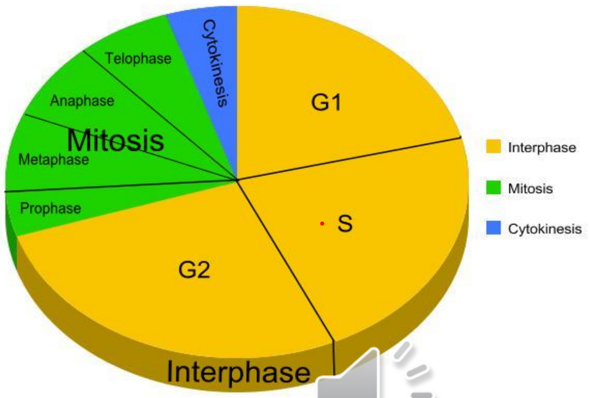
1- Cell growth

2-Synthesis of new organelles

3-Protein synthesis is high

4- Production enzymes for S phase

5- Has checkpoint



Interphase

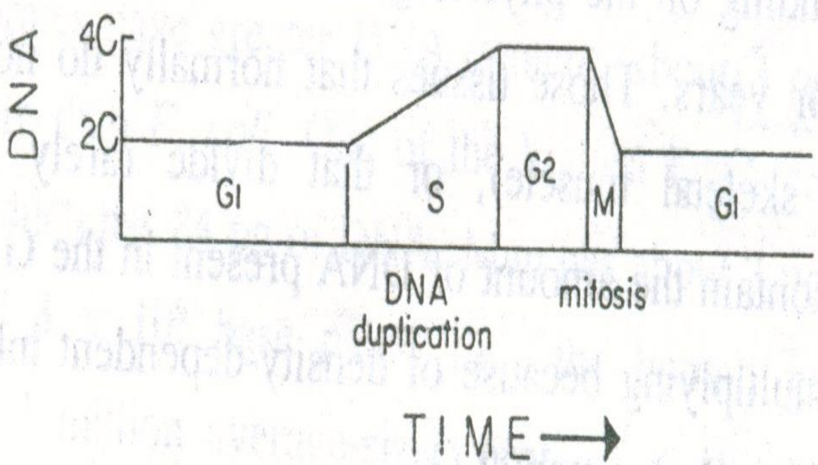
- **Synthetic stage**(S) is the period of DNA synthesis, during s phase. The interval between G1 and G2
- Protein synthesis is low
- Histones are produced
- **Second growth stage** (G_2), the interval between the end of DNA synthesis and the start of mitosis. G2 stage characterized by
- 1- Growth for preparation of mitosis
- 2-Synthesis of microtubules

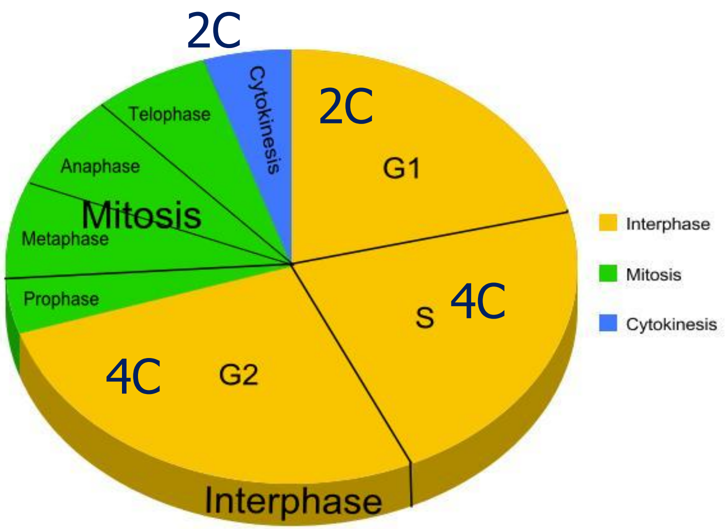
Interphase

The amount of DNA present in the original diploid cell (2C)

G_2 (4C)

During mitosis the daughter cells again enter the G_1 period and have a DNA content equivalent to 2C





Changes in DNA content during different period of life cycle

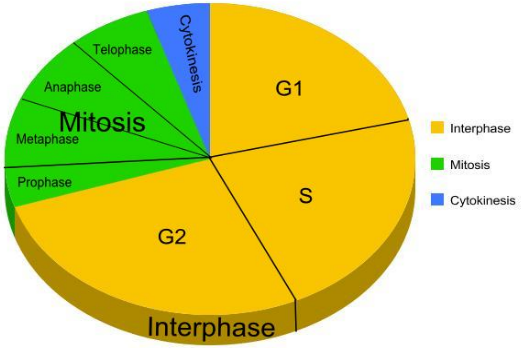
The duration of the cell cycle varies greatly from one cell to another.

For a mammalian cell growing in culture with a generation time of 16 hours,

The different periods would be as follows:

G_1 = 5 hours, S = 7 hours,

G_2 = 3 hours, mitosis = 1 hour.

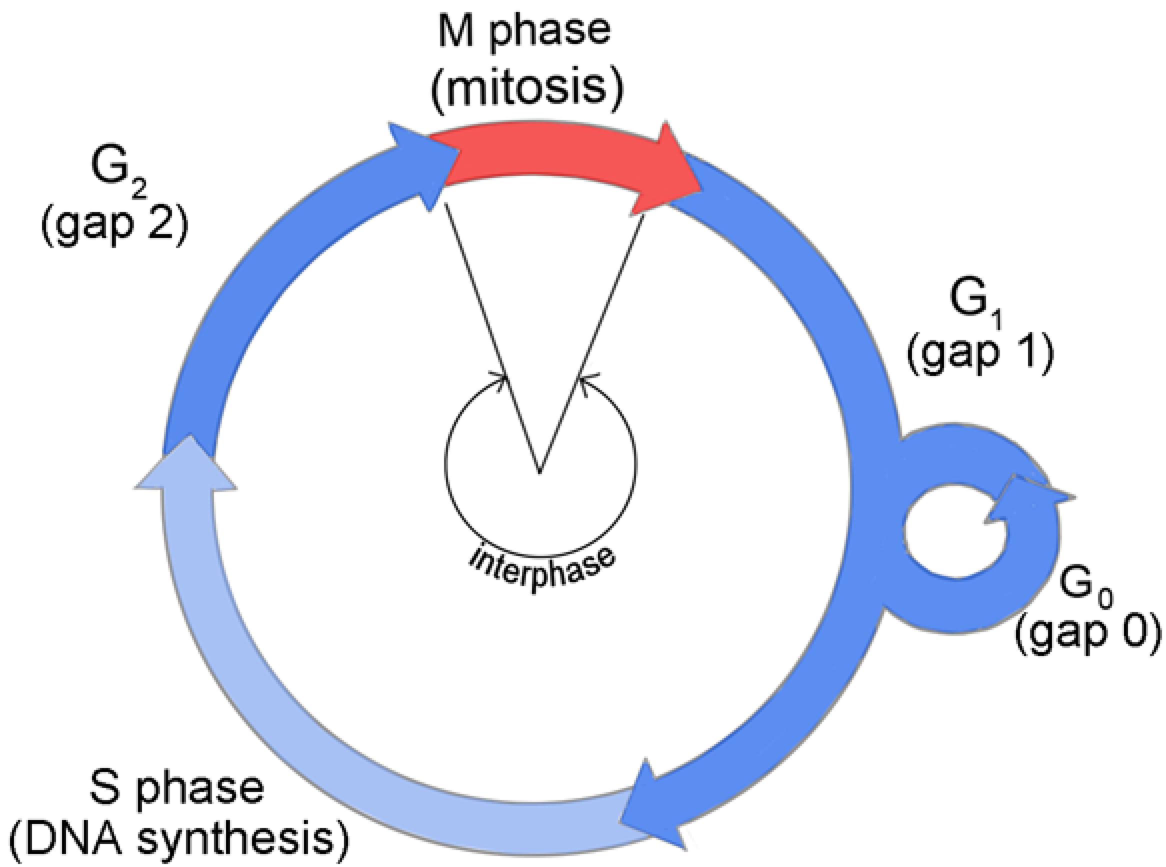


The S, G_2 and mitotic periods are relatively constant in the cells of the same organism.

The G_1 period is the most variable in length. Depending on the physiological condition of the cells, it may last days, months, or years.

Those tissues that normally do not divide (such as nerve cells or skeletal muscle), or that divide rarely (e.g., circulating lymphocytes), contain the amount of DNA present in the G_1 period.

The regulation of the duration of the cell cycle occurs primarily by arresting it at a specific point of G1, and the cell in the arrested condition is said to be in the G0 state.


M phase
(mitosis)
G2
(gap 2)
G1
(gap 1)
G0
(gap 0)
interphase
S phase
(DNA synthesis)

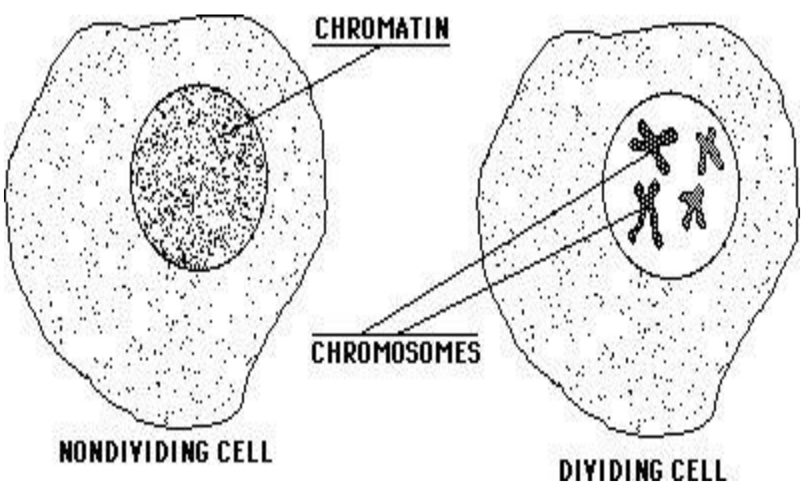
Genetic material

DNA + protein
Morphology

Interphase

Chromatin

indistinct

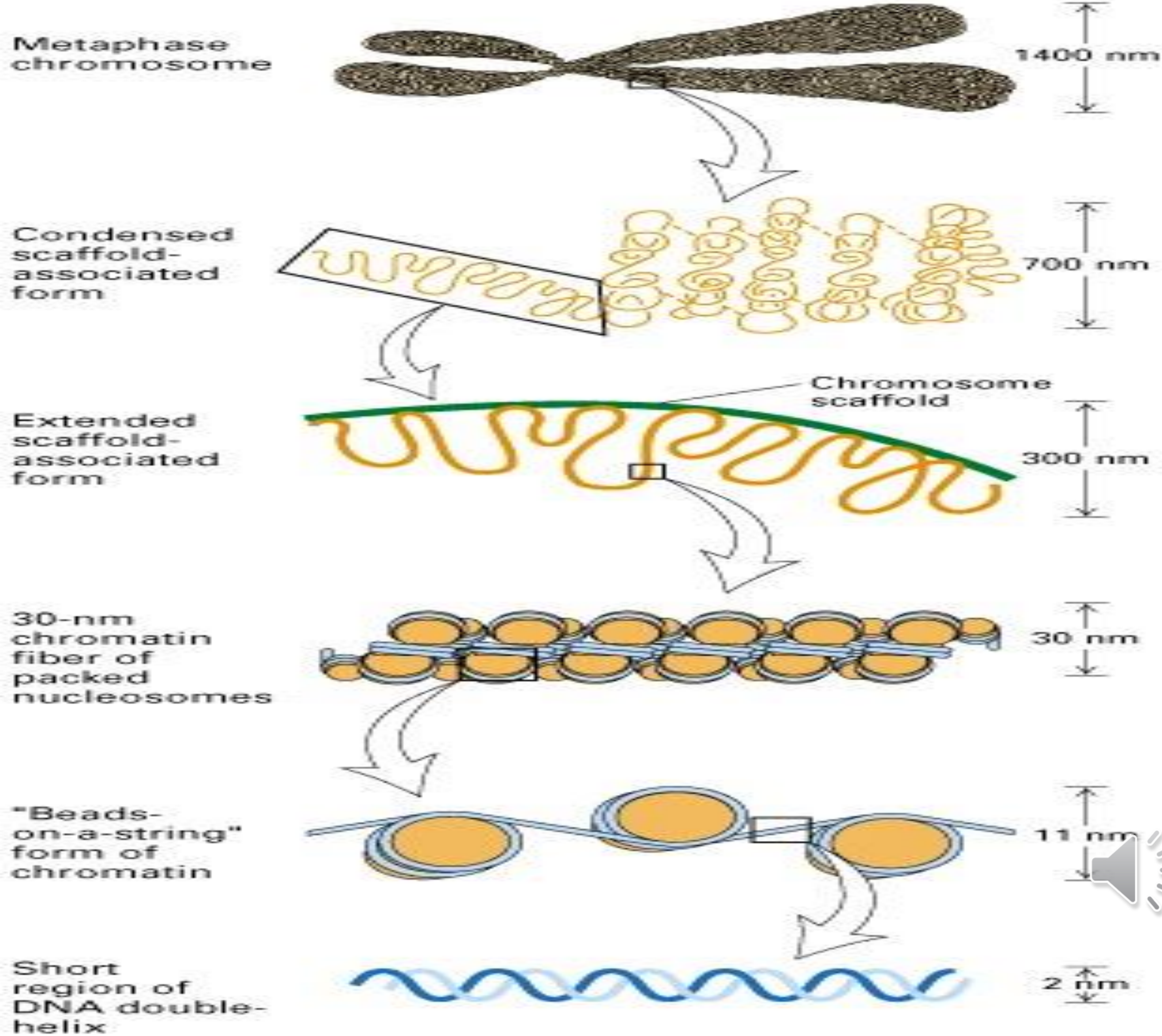


During division

Chromosome

During cell division become quite distinct in microscopical preparations ,

when they accept dyes intensely, and hence the ***name chromosome*** (chromo = coloured; somes = bodies).

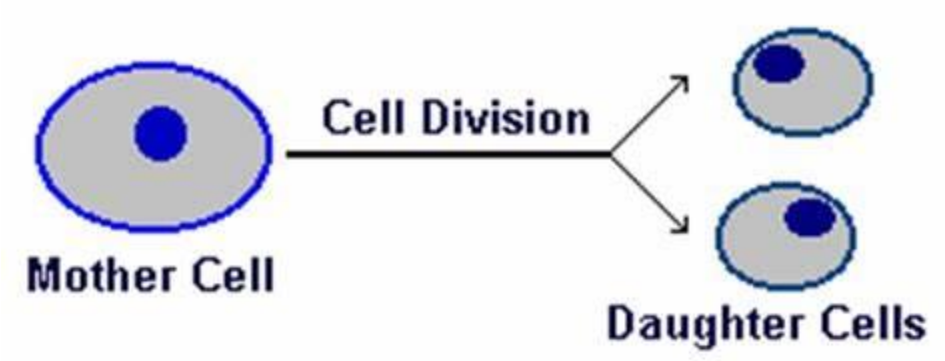

Metaphase chromosome
1400 nm
Condensed scaffold-associated form
700 nm
Chromosome scaffold
Extended scaffold-associated form
300 nm
30-nm chromatin fiber of packed nucleosomes
30 nm
"Beads-on-a-string" form of chromatin
11 nm
Short region of DNA double-helix
2 nm

The C-Value

The amount in picograms of DNA contained within a Diploid somatic cell of eukaryotic organism

Or one half of this amount in a haploid nucleus (gamete)

The demonstration that nuclei contain a constant amount of DNA was a landmark in cell biology because it suggested that DNA was the genetic material, and that genetic information is not lost during the differentiation of the various somatic tissues.

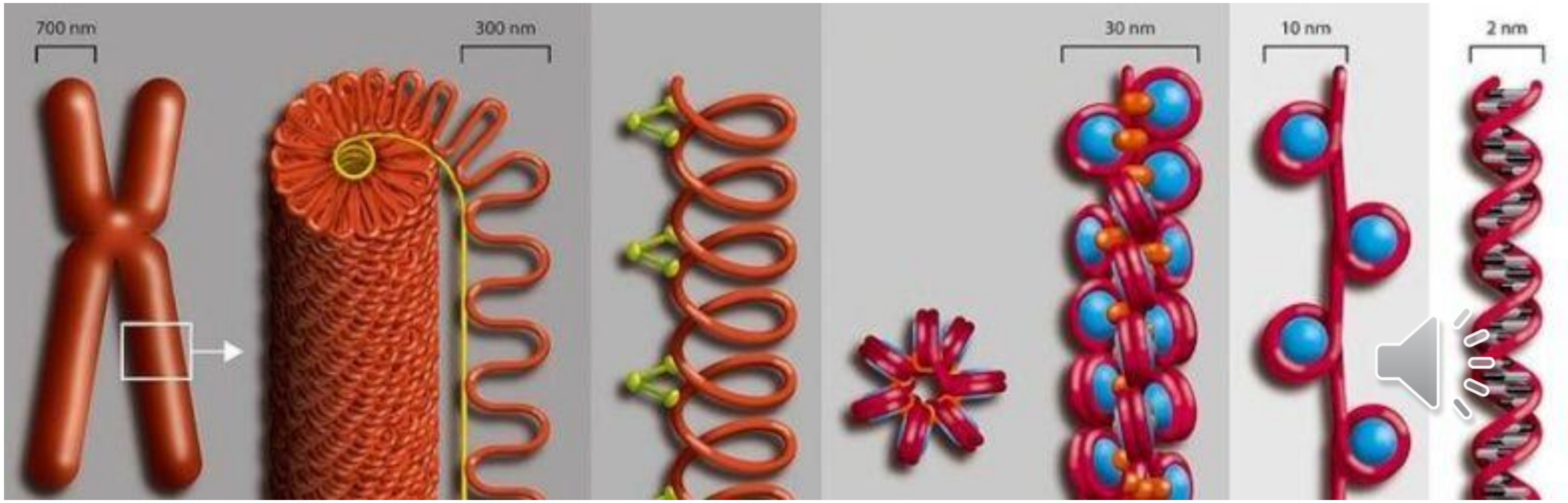


- All the cells in an organism contain the same DNA content (2C) provided that they are diploid.

- Gametes are haploid and therefore have half the DNA content (1C).

- Some tissues, like liver, contain occasional cells that are polyploid and their nuclei have a higher DNA content (4C or 8C)

- 1 pg of DNA = 31 cm of DNA,
  =174 cm of DNA human diploid cells.
- The DNA content in the 46 human chromosomes has been estimated by cytophotometry.
- The largest chromosome (1), which is 10 µm long, should accommodate about 7.2 cm of DNA
- The DNA is compacted about 7000-fold in a metaphase chromosome.




THANK
YOU